DECLARATION

We hereby declare that the work presented in this report entitled "**Language Agnostic AI Chatbot**" was carried out by us. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

We have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments and results that are not our original contribution. We have used quotation marks to identify verbatim sentences and given credit to the original authors/sources.

We affirm that no portion of our work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, we shall be fully responsible and answerable.

ADHEEP OJHA	Roll No. - 2405250139001	_____
MONIKA SINGH	Roll No. - 2305250130034	_____
KUMARI KIRAN	Roll No. - 2305250130027	_____
ADITI YADAV	Roll No. - 2305251540003	_____

(Candidates' Signature)

Date: _____

Department of Information Technology

CERTIFICATE

This is to certify that the Project Report entitled "**Language Agnostic AI Chatbot**" submitted by **Adheep Ojha, Monika Singh, Kumari Kiran and Aditi Yadav** in partial fulfilment of the requirement for the award of **Bachelor of Technology in Information Technology** from Buddha Institute of Technology, Gorakhpur, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh, represents the work carried out by the students under my supervision. The project embodies the result of original work and studies carried out by the students themselves, and the contents of the project do not form the basis for the award of any other degree to the candidates or to anybody else.

_____

Miss Purva Rai
(Assistant Professor)
Department of Information Technology
Buddha Institute of Technology, GIDA, Gorakhpur

_____

Mr. Ranjeet Rai
(Assistant Professor & Head)
Department of Information Technology
Buddha Institute of Technology, GIDA, Gorakhpur

Date: _____

ABSTRACT

Language remains one of the biggest barriers to the use of digital services in India and in many other multilingual regions. Most chatbots and voice assistants support only a few languages, treat typed and spoken interaction as separate products, and offer little help with everyday tasks such as filling in online forms. This project presents a **Language Agnostic AI Chatbot** that allows a person to communicate in the language they are most comfortable with, whether by typing, by speaking into a microphone, or by calling over the telephone, without selecting a language in advance or switching between modes.

The system uses large language models to understand and answer in the language used by the person, including code-mixed speech such as Hinglish. It brings together two independently developed components. **Voicera** is a low-latency voice agent built as a pipeline of voice activity detection, speech-to-text, a streaming language model and text-to-speech, with sentence-level streaming to reduce the time to first audio and a telephony interface for phone calls. **AutoForm AI** is a browser extension that reads the questions of an online form and fills in answers generated by AI. The report covers requirement analysis, system architecture, the implementation of each module, their integration into a single platform, and testing across languages and input modes, together with a user evaluation of usability, response quality and perceived delay. The result is a practical and accessible assistant that lowers language and interface barriers for users.

Keywords: Language Agnostic Chatbot, Multilingual NLP, Large Language Models, Voice AI, Speech Recognition, Text-to-Speech, Browser Extension, Form Automation.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide, **Miss Purva Rai** (Assistant Professor, Department of Information Technology), for her valuable guidance, constant encouragement and support throughout the course of this project.

We are thankful to **Mr. Ranjeet Rai**, Head of the Department of Information Technology at Buddha Institute of Technology, Gorakhpur, for his valuable advice and for providing the facilities needed to carry out this work.

We convey our sincere thanks to all the faculty members and technical staff of the department for their help and encouragement. We also thank the open-source community, whose tools and libraries made it possible to build and test the system, and the friends and classmates who took part in testing and gave us honest feedback.

We are deeply grateful to our parents for their constant encouragement and enduring support, which was essential for this success. Once again, we convey our gratitude to all those who have directly or indirectly helped us in this work.

ADHEEP OJHA	Roll No. - 2405250139001
MONIKA SINGH	Roll No. - 2305250130034
KUMARI KIRAN	Roll No. - 2305250130027
ADITI YADAV	Roll No. - 2305251540003

TABLE OF CONTENTS

	Page No.
Declaration	i
Certificate	ii
Abstract	iii
Acknowledgement	iv
Table of Contents	v
List of Tables	vii
List of Figures	viii
List of Abbreviations	ix
Chapter 1 INTRODUCTION	**1-8**
1.1 Background	1
1.2 Problem statement	3
1.3 Objectives of the project	4
1.4 Scope and limitations	5
1.5 Organization of the report	7
Chapter 2 LITERATURE REVIEW	**9-18**
2.1 Overview	9
2.2 Multilingual chatbots and large language models	9
2.3 Voice AI: voice activity detection, speech recognition and synthesis	12
2.4 Browser-based form automation	15
2.5 Research gap	17
Chapter 3 SYSTEM ANALYSIS AND DESIGN	**19-32**
3.1 Overview	19
3.2 Requirement analysis	19
3.2.1 Functional requirements	20
3.2.2 Non-functional requirements	22
3.3 System architecture	23
3.4 Language-agnostic processing pipeline	26
3.5 Data flow diagrams	28
3.6 Database design	30
Chapter 4 IMPLEMENTATION	**33-52**
4.1 Technology stack	33
4.2 Text chatbot module	35
4.3 Voicera: the voice agent module	38
4.3.1 Voice activity detection	39

Page No.

4.3.2 Speech-to-text 40
4.3.3 Streaming language model response 42
4.3.4 Text-to-speech 43
4.3.5 Telephony integration 44
4.4 AutoForm AI: the form-filling module 46
4.5 Integration of the modules 49
4.6 Deployment 51
Chapter 5 TESTING AND RESULTS 53-64
5.1 Test setup 53
5.2 Multilingual response testing 54
5.3 Voice pipeline latency 57
5.4 Form auto-fill evaluation 59
5.5 User evaluation 61
5.6 Discussion 63
Chapter 6 CONCLUSION AND FUTURE SCOPE 65-67
6.1 Conclusions 65
6.2 Limitations 66
6.3 Future scope 66
REFERENCES 68
APPENDICES 70
Appendix I Project structure and setup 70
Appendix II User evaluation questionnaire 72

LIST OF TABLES

Table No.	Description	Page No.
Table 2.1	Comparison of existing multilingual chatbot and voice assistant solutions	16
Table 3.1	Functional requirements of the system	20
Table 3.2	Non-functional requirements of the system	22
Table 4.1	Technology stack used in the project	34
Table 4.2	Speech-to-text options considered for Hindi and Hinglish	41
Table 5.1	Languages and test cases used for evaluation	55
Table 5.2	Time-to-first-audio measurements of the voice pipeline	58
Table 5.3	Form auto-fill results	60
Table 5.4	User evaluation summary	62

LIST OF FIGURES

Figure No.	Description	Page No.
Figure 3.1	Overall architecture of the Language Agnostic AI Chatbot	24
Figure 3.2	Language-agnostic processing pipeline	27
Figure 3.3	Level-0 data flow diagram	29
Figure 3.4	Database schema	31
Figure 4.1	Chat interface of the chatbot	36
Figure 4.2	Voicera pipeline: VAD, STT, LLM and TTS stages	38
Figure 4.3	Telephony call flow	45
Figure 4.4	AutoForm AI workflow	47
Figure 5.1	Time-to-first-audio comparison	58
Figure 5.2	User evaluation ratings	62

LIST OF ABBREVIATIONS

Abbreviation	Full form
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BYOK	Bring Your Own Key
DFD	Data Flow Diagram
JSON	JavaScript Object Notation
LLM	Large Language Model
NLP	Natural Language Processing
STT	Speech-to-Text
TTFA	Time to First Audio
TTS	Text-to-Speech
UI	User Interface
VAD	Voice Activity Detection